

数学作业纸

科目_____

班级:_____

姓名:_____

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3.1 b $-F_{N3} + 15kN = 0$

$F_{N3} = 15kN$

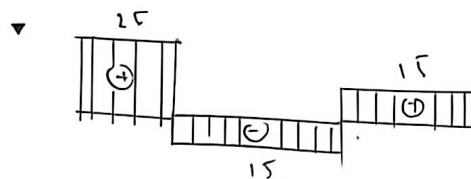
$-F_{N2} - 30kN + 15kN = 0$

$F_{N2} = -15kN$

$-F_{N1} + 40kN - 30kN + 15kN = 0$

$-F_{N1} + 25kN = 0$

$F_{N1} = 25kN$



3.2. B C ~~段~~. $F_{N1} = A \Delta x_1 = 16 \times 10^{-2} \times 24 x_1 = 3.84 x_1$ (kN) ($0 \leq x_1 \leq 1m$)

$\frac{1}{2} x_1 = 1m$

$F_{NB}^{\text{上}} = A \Delta x_1 = 16 \times 10^{-2} \times 24 \times 1 = 3.84kN$ (kN)

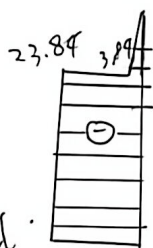
B A ~~段~~

$F_{N2} = F + A \Delta x_2 = F + 16 \times 10^{-2} \times 24 x_2 = 20 + 3.84 x_2$ (kN) ($1 \leq x_2 \leq 4m$)

$\frac{1}{2} x_2 = 1m$

$F_{NB}^{\text{下}} = 20 + A \Delta x_2 = 20 + 16 \times 10^{-2} \times 24 \times 1 = 23.84kN$ (kN)

$F_{NA}^{\text{下}} = 20 + 2 \Delta x_2 = 20 + 16 \times 10^{-2} \times 24 \times 4 = 35.36kN$ (kN)



35.36

11 $F_{N2} = F$ $F_{N1} = F + 2F = 3F$

$\Delta L = \Delta L_1 + \Delta L_2 = \frac{F_{N1} \cdot L_1}{E_1 \cdot A_1} + \frac{F_{N2} \cdot L_2}{E_2 \cdot A_2}$

$0.68 = \frac{3F \cdot 600}{(200 \times 10^3) \times 2000} + \frac{F \cdot 400}{(100 \times 10^3) \times 1000} = \frac{45F + 40F}{10^6 \pi} = \frac{85F}{10^6 \pi}$

$F = \frac{85 \times 0.68 \times 10^6 \pi}{85} = 25.1kN$

$\sigma_1 = \frac{F_{N1}}{A_1} = \frac{3F}{2000} = \frac{24000 \pi}{2000} = 120MPa$

$\sigma_2 = \frac{F_{N2}}{A_2} = \frac{F}{1000} = 25.1MPa$

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$$\sigma_{\max} = 120 \text{ MPa}.$$

$$3.30 \cdot N_1 + N_2 + N_3 = \bar{F}.$$

$$\sum M_A = 0 \Rightarrow \bar{F} \cdot 2a - N_2 \cdot a - N_3 \cdot 2a = 0.$$

$$N_2 + 2N_3 = 2\bar{F}.$$

$$\delta_2 = \frac{\Delta_1 + \Delta_3}{2} \quad \delta_1 + \delta_3 = 2\delta_2.$$

$$\delta_1 = \frac{N_1 L}{EA} \quad \delta_2 = \frac{N_2 L}{EA} \quad \delta_3 = \frac{N_3 L}{EA}.$$

$$\frac{N_1 L}{EA} + \frac{N_3 L}{EA} = 2 \left(\frac{N_2 L}{EA} \right).$$

$$N_1 + N_3 = 2N_2.$$

$$\begin{cases} N_1 + N_2 + N_3 = \bar{F} \\ N_2 + 2N_3 = 2\bar{F} \\ N_1 - 2N_2 + N_3 = 0 \end{cases} \Rightarrow \begin{cases} N_1 = -\frac{\bar{F}}{6} \\ N_2 = \frac{\bar{F}}{3} \\ N_3 = \frac{5}{6}\bar{F} \end{cases}.$$